

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 3

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1. TITLE

Design a Responsive Webpage Using HTML5 and CSS3

2. AIM

To design and develop a modern, responsive webpage using **HTML5 and CSS3** with a navigation bar, hero section, about section, services section, contact form, and footer.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML5 webpage.
 - To design a webpage using HTML5 semantic elements.
 - To apply CSS3 styling to HTML elements.
 - To create a responsive navigation bar.
 - To design an attractive hero section using CSS.
 - To create service cards using CSS Flexbox.
 - To design and style a contact form.
 - To implement hover effects and smooth scrolling.
 - To make the webpage responsive using CSS media queries.
 - To understand the separation of HTML structure and CSS presentation.
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4. THEORY

HTML5 (HyperText Markup Language 5) is used to define the structure and content of a webpage. It provides elements such as `<header>`, `<nav>`, `<section>`, `<form>`, and `<footer>` for organizing webpage content.

CSS3 (Cascading Style Sheets 3) is used to control the presentation and appearance of HTML elements. CSS can be used to define colors, fonts, spacing, layouts, backgrounds, borders, animations, and responsive behavior.

In this experiment, HTML5 is used to create the structure of a website called **MyWebsite**, while an external CSS file named `styles.css` is used for styling.

The webpage consists of:

- **Header:** Contains the website logo and navigation links.
- **Hero Section:** Contains the main welcome message and Learn More button.
- **About Section:** Provides information about the website.
- **Services Section:** Displays three service cards.
- **Contact Section:** Contains a form for entering user information.  **Footer:** Displays copyright information.

CSS3 features such as **Flexbox, gradients, transitions, hover effects, sticky positioning, box shadows, and media queries** are used to improve the appearance and responsiveness of the webpage.

The basic working flow is:

HTML Structure → CSS Styling → Browser Rendering → Responsive Webpage

5. ALGORITHM / PROCEDURE

1. Create a new project folder for the webpage.
2. Create an HTML file named `index.html`.
3. Create an external CSS file named `styles.css`.
4. Define the HTML5 document structure using `<!DOCTYPE html>`.
5. Add the webpage title and link the external CSS file.
6. Create a header containing the website logo and navigation links.
7. Create a hero section with a heading, description, and button.
8. Create an About section containing information about the website.
9. Create a Services section containing three service cards.
10. Create a Contact section containing a form with name, email, and message fields.
11. Create a footer containing copyright information.
12. Use CSS to style the webpage elements.
13. Apply Flexbox to arrange navigation elements and service cards.
14. Apply colors, spacing, typography, gradients, borders, and shadows using CSS.
15. Add hover effects to navigation links, buttons, and service cards.
16. Add smooth scrolling using the `scroll-behavior` property.

17. Use `position: sticky` to keep the header visible while scrolling.
 18. Implement responsive design using the `@media` rule.
 19. Save both HTML and CSS files in the same project folder.
 20. Open `index.html` in a web browser and verify the complete webpage output.
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6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Webpage</title>
  <link rel="stylesheet" href="styles.css">
</head>

<body>

  <!-- Header -->
  <header>
    <div class="logo">MyWebsite</div>

    <nav>
      <a href="#home">Home</a>
      <a href="#about">About</a>
      <a href="#services">Services</a>
      <a href="#contact">Contact</a>
    </nav>
  </header>

  <!-- Hero Section -->
  <section class="hero" id="home">
    <div class="hero-content">
      <h1>Welcome to My Website</h1>
      <p>
        A simple and modern webpage designed using HTML5 and CSS3.
      </p>
      <a href="#about" class="btn">Learn More</a>
    </div>
  </section>

  <!-- About Section -->
```

```
<section class="about" id="about">
  <h2>About Us</h2>
  <p>
    We create clean, responsive and user-friendly websites using
    modern web technologies.
  </p>
</section>

<!-- Services Section -->
<section class="services" id="services">
  <h2>Our Services</h2>

  <div class="service-container">

    <div class="card">
      <h3>Web Design</h3>
      <p>
        Attractive and responsive website designs for all
        types of devices.
      </p>
    </div>

    <div class="card">
      <h3>Development</h3>
      <p>
        Modern websites developed using HTML5, CSS3 and
        JavaScript.
      </p>
    </div>

    <div class="card">
      <h3>Support</h3>
      <p>
        Reliable maintenance and support for your website.
      </p>
    </div>

  </div>
</section>

<!-- Contact Section -->
<section class="contact" id="contact">
  <h2>Contact Us</h2>

  <form>
    <input type="text" placeholder="Your Name" required>
```

```
        <input type="email" placeholder="Your Email" required>
placeholder="Your Message" rows="5"></textarea>
        <button type="submit">Send Message</button>
    </form>
</section>
```

```
<!-- Footer -->
<footer>
    <p>&copy; 2026 MyWebsite. All Rights Reserved.</p>
</footer>
```

```
</body>
</html>
```

```
styles.css * {    margin: 0;
padding: 0;    box-sizing:
border-box;    scrollbehavior:
smooth;
}
```

```
body {    font-family: Arial, sans-serif;
line-height: 1.6;
color: #333;    background-color:
#f5f5f5;
}
```

```
/*    Header    */    header
{    background-color: #222;    color:
white;    display: flex;
justifycontent: space-between;
alignitems: center;    padding: 20px
8%;    position: sticky;    top: 0;
z-index: 1000;
}
```

```
.logo {    font-size: 26px;
font-weight:
bold; }
```

```
nav {    display:
flex;    gap:
25px;
}
```

```
nav a { color: white; text-decoration:
none; font-size: 16px; transition:
0.3s;
}
```

```
nav a:hover { color:
#00adb5;
}
```

```
/* Hero Section */ .hero {
height: 90vh; display: flex;
justifycontent: center;
align-items: center; text-
align: center; background:
linear-gradient(
135deg,
#00adb5,
#393e46
);
color: white; padding:
20px;
}
```

```
.hero-content { maxwidth:
700px;
}
```

```
.hero h1 { font-size: 50px;
margin-bottom:
20px;
}
```

```
.hero p { font-size: 20px;
margin-bottom:
30px;
}
```

```
.btn { display: inlineblock;
background-color: white; color:
#222; padding: 12px
25px;
border-radius: 5px;
textdecoration: none;
fontweight: bold;
transition:
0.3s;
```

```
}
```

```
.btn:hover { background-color:
#222; color: white;
}
```

```
/* Common Section Styles */ section
{ padding: 70px
8%; text-align: center;
}
```

```
section h2 { font-size: 35px;
margin-bottom:
20px;
}
```

```
/* About */ .about { background-
color: white;
}
```

```
.about p { maxwidth:
700px; margin: auto;
font-size: 18px;
}
```

```
/* Services */ .services { background-color:
#f5f5f5;
}
```

```
.service-container { display:
flex; justify-content: center;
gap:
25px; margin-top:
35px; flex-wrap: wrap;
}
```

```
.card { background-color: white; width: 300px;
padding: 30px; border-radius:
10px; box-shadow: 0 5px 15px rgba(0, 0,
0, 0.1);
transition: 0.3s;
}
```

```
.card:hover { transform:
translateY(-8px);
}
```

```
.card h3 {  color:
#00adb5;
marginbottom: 15px;
fontsize: 23px; }

/* Contact */ .contact {
backgroundcolor: white;
}

form {  max-width: 600px;
margin: 30px auto 0;
display: flex;  flex-direction:
column;
gap: 15px; }

input, textarea {  width:
100%;  padding: 14px;
border: 1px solid #ccc;
borderradius: 5px;
font-size:
16px;  font-family:
inherit; }

textarea {
resize: vertical;
}

button {  padding: 14px;
border: none;  border-radius: 5px;
background-color: #00adb5;
color: white;  font-size: 17px;
cursor: pointer;  transition: 0.3s;
}

button:hover {  backgroundcolor:
#008b91;
}

/* Footer */ footer {
background-color: #222;
color: white;  text-align: center;
padding: 20px;
}

/* Responsive Design */ @media (max-width: 768px) {
```

```

    header {      flex-
direction: column;
gap: 15px;
    }

    nav {      gap: 15px;      flex-wrap:
wrap;
justify-content: center;
    }

    .hero h1 {      font-size:
36px;
    }

    .hero p {
        font-size: 17px;
    }

    section {
        padding: 50px 5%;
    }

    .service-container {      flexdirection:
column;      align-items:
center;
    }

    .card {      width:
100%;
        max-width: 350px;
    }
}

```

7. CODE EXPLANATION

Code	Explanation
<code><!DOCTYPE html></code>	Declares the document as an HTML5 document.
<code><meta charset="UTF-8"></code>	Defines the character encoding as UTF-8.
<code><meta name="viewport"...></code>	Makes the webpage responsive on different screen sizes.
<code><title></code>	Specifies the title displayed in the browser tab.
<code><link rel="stylesheet"...></code>	Connects the external CSS file with the HTML document.

<code><header></code>	Defines the header section of the webpage.
Code	Explanation
<code><nav></code>	Contains the main navigation links.
<code><section></code>	Defines individual sections of the webpage.
<code>id</code>	Provides a unique identifier for internal navigation.
<code><div></code>	Groups related HTML elements together.
<code><a></code>	Creates hyperlinks and navigation links.
<code><form></code>	Creates the contact form.
<code><input></code>	Accepts user input such as name and email.
<code><textarea></code>	Provides a multi-line input area for messages.
<code><button></code>	Creates the form submission button.
<code><footer></code>	Defines the footer section.
<code>display: flex</code>	Creates a flexible layout for arranging elements.
<code>justify-content</code>	Controls the horizontal alignment of flex items.
<code>align-items</code>	Controls the vertical alignment of flex items.
<code>gap</code>	Adds spacing between flex items.
<code>background-color</code>	Sets the background color of an element.
<code>background: lineargradient()</code>	Creates a gradient background for the hero section.
<code>border-radius</code>	Creates rounded corners.
<code>box-shadow</code>	Adds a shadow effect to service cards.
<code>transition</code>	Creates smooth visual transitions.
<code>:hover</code>	Applies styles when the mouse pointer moves over an element.
<code>position: sticky</code>	Keeps the header fixed within its scrolling position.
<code>scroll-behavior: smooth</code>	Provides smooth scrolling between page sections.
<code>@media</code>	Defines responsive styles for different screen sizes.
<code>flex-wrap</code>	Allows flex items to move to the next line when required.

8. TEST CASES AND EXECUTION DATA

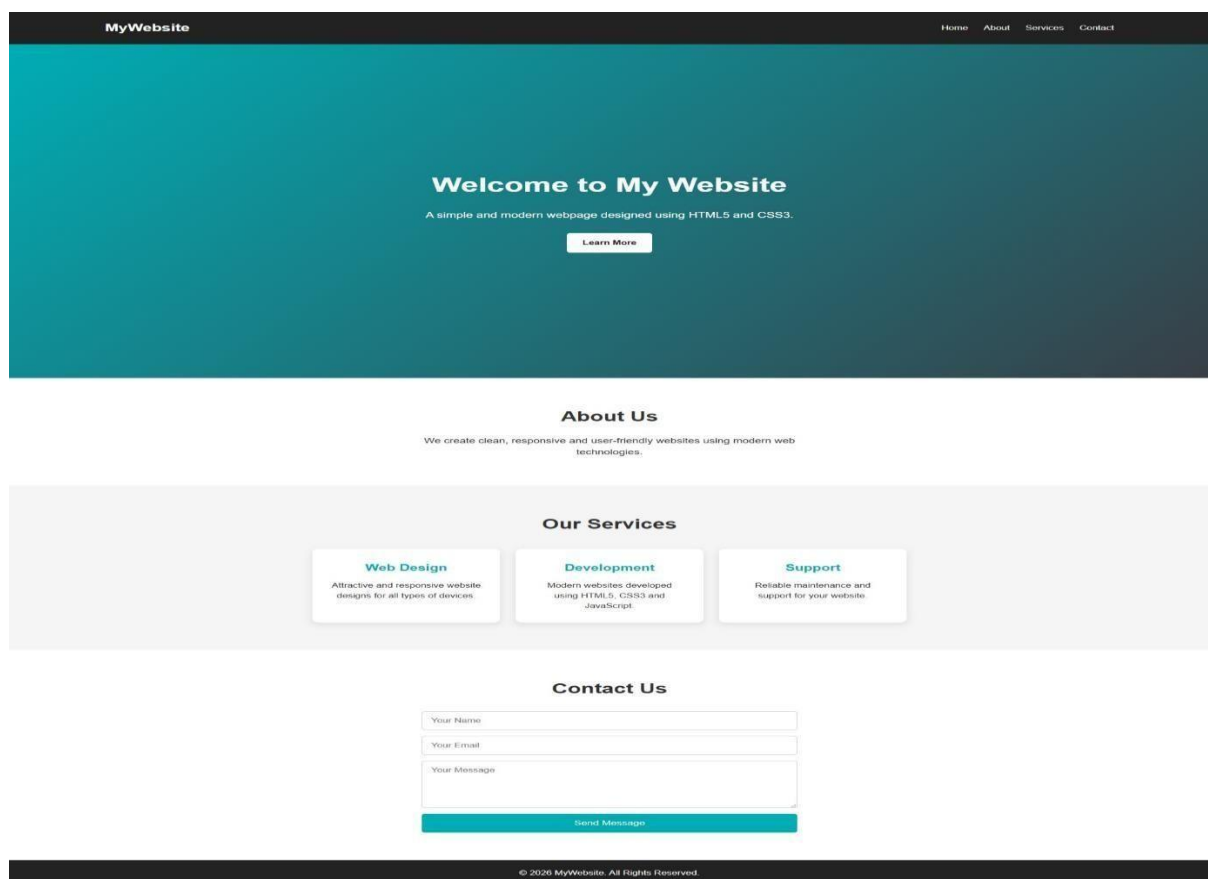
The webpage was executed in a web browser and tested for navigation, layout, styling, form controls, hover effects, and responsive behavior.

Test Case	Test / Input	Expected Result	Actual Result	Status
TC-01	Open <code>index.html</code>	Complete webpage should be displayed	Webpage displayed successfully	Pass
TC-02	Click Home	Page should navigate to Home section	Home section displayed	Pass
TC-03	Click About	Page should navigate to About section	About section displayed	Pass
TC-04	Click Services	Page should navigate to Services section	Services section displayed	Pass
TC-05	Click Contact	Page should navigate to Contact section	Contact section displayed	Pass
TC-06	Click Learn More	Page should navigate to About section	About section displayed	Pass
TC-07	Enter name and email in contact form	Input values should be accepted	Values accepted successfully	Pass
TC-08	Enter message in textarea	Message should be accepted	Message entered successfully	Pass
TC-09	Hover over navigation links	Hover effect should be displayed	Hover effect displayed	Pass
TC-10	Hover over service cards	Card should move upward smoothly	Hover animation displayed	Pass
TC-11	Resize browser to mobile width	Layout should adapt to smaller screen	Responsive layout displayed	Pass
TC-12	Leave required form fields empty	Browser should display validation	Required-field validation displayed	Pass

9. OUTPUT

Output: Complete Responsive Webpage

The following screenshot shows the complete execution of the **HTML5 and CSS3 webpage** in the web browser. It contains the header and navigation bar, hero section, About section, Services section, Contact form, and footer.



10. RESULT

Thus, a modern and responsive webpage was successfully designed and developed using **HTML5 and CSS3**. The webpage contains a navigation bar, hero section, About section, service cards, contact form, and footer. CSS3 features such as Flexbox, gradients, transitions, hover effects, sticky positioning, box shadows, and media queries were successfully implemented. The webpage was executed and verified successfully in a web browser.

